A Computer-Assisted Proof of Global Strict Convexity and Disk-Like Global Sections for the Earth–Moon PCR3BP

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Abstract

We give a computer-assisted proof of global strict convexity for the bounded lunar component of the planar circular restricted three-body problem at the physical Earth–Moon mass ratio. We use the Hamiltonian convention $\Sigma_c = H^{-1}(-c)$, with heavy mass $M = 0.987849414390376$ and small mass parameter $\mu = 0.012150585609624$. The first critical value is enclosed by

$$c_{L_1} \in [1.5941705588746069, 1.5941705588746333],$$

and the theorem covers every regular subcritical energy $c > c_{L_1}$. After Levi–Civita regularization, the level is written as $|u|^2 = F_c(x, y)$. A global exact antipodal symplectic shear generated by $\frac{1}{2}p_1p_2(p_1^2 - p_2^2)$ converts the tangential convexity test into positivity of an explicit 3×3 interval matrix. The regular range is divided into a critical bridge, five compact atlas blocks, a lower finite gap, ten Kepler blocks, and an analytic high-energy tail. All adjacent intervals meet exactly and every certified lower bound is positive; the global minimum is $1.2482059830176695 \times 10^{-10}$, attained in block M18A. Consequently the regularized lunar component is dynamically convex. Birkhoff's retrograde orbit is a 2-unknot with rational self-linking number $-1/2$ in $\mathbb{R}P^3$, and therefore binds a rational open book with disk-like global pages. The release includes source code, interval certificates, manifests, SHA-256 hashes, cross-compiler replays, and scripts for complete verification.

Keywords. planar circular restricted three-body problem; Earth–Moon system; Birkhoff conjecture; global surface of section; Levi–Civita regularization; strict convexity; interval arithmetic; computer-assisted proof; Reeb dynamics.

MSC 2020. 37J46, 37J55, 53D10, 70F07, 65G20.

1 Introduction

The planar circular restricted three-body problem (PCR3BP) is a classical two-degree-of-freedom Hamiltonian system in celestial mechanics. Birkhoff proved the existence of a retrograde periodic orbit in each bounded component below the first Lagrange value and proposed that this orbit should bound a disk-like global surface of section [1]. Perturbative constructions were obtained for small mass parameters [2], while contact and symplectic methods established restricted-contact-type and global-section results in additional parameter ranges [3, 4]. The lens-space theorem of Hryniewicz and Salomão identifies a direct route from dynamical convexity and the topology of the retrograde orbit to a rational disk-like global section on $\mathbb{R}P^3$ [5].

Recent validated-numerics work has developed computational symplectic tools for the restricted problem [6]. Most closely related to the present result, Liu and Salomão prove Birkhoff's retrograde-orbit conjecture for mass ratios sufficiently close to $1/2$ and for all energies below the first Lagrange value, using finite-energy foliations and a critical-convexity analysis [7]. They also prove the full retrograde-orbit conjecture in Hill's lunar limit [8]. The present paper treats the physical Earth–Moon mass ratio and the entire regular subcritical energy range. The proof is organized around a global convexity certificate for the Levi–Civita lift. The central difficulty is that one coordinate system and one numerical decomposition are not uniformly efficient from the first critical level to the Kepler limit. We therefore use a rigorously matched collection of critical, compact, finite-energy, and asymptotic certificates.

The contribution has four parts. First, we derive the exact Levi–Civita level equation and a global polynomial symplectic shear. Second, we identify the matrix tested in the interval code with the Hessian of the defining function restricted to the tangent space. Third, we cover the full regular range by certificates with exact interfaces. Fourth, we apply standard contact-topological results to obtain the disk-like global section bound by the retrograde orbit.

2 Hamiltonian, energy convention, and lunar component

Let $q = (q_1, q_2)$ and $p = (p_1, p_2)$. We use

$$H(q, p) = \frac{1}{2}|p|^2 + q_1 p_2 - q_2 p_1 - \frac{1-\mu}{|q + (\mu, 0)|} - \frac{\mu}{|q - (1-\mu, 0)|}. \tag{1}$$

The energy hypersurface is

$$\Sigma_c = H^{-1}(-c). \tag{2}$$

Thus the astronomical Jacobi constant is $C_J = -2H = 2c$ on Σ_c. For the Earth–Moon parameter,

$$M = 1 - \mu = 0.987849414390376, \qquad \mu = 0.012150585609624.$$

The first critical value satisfies

$$c_{L_1} \in [1.5941705588746069, 1.5941705588746333],$$
$$C_J(L_1) \in [3.188341117749214, 3.1883411177492667].$$

Hamiltonian energies below the first Lagrange value correspond to $c > c_{L_1}$ in (2). The critical level itself contains the equilibrium and is excluded from the regular theorem.

We translate the light primary to the origin by $Q = q - (M, 0)$. The bounded physical component under consideration is the lunar component surrounding $Q = 0$.

3 Levi–Civita regularization

Set

$$Q_1 = -2(x^2 - y^2), \qquad Q_2 = -4xy, \qquad s = x^2 + y^2.$$

Then $|Q| = 2s$ and

$$d = |Q + (1, 0)|^2 = 1 - 4(x^2 - y^2) + 4s^2.$$

Let $w = DQ(x, y)^T p$ and define

$$b(x, y) = (-(2s + M)y, (2s - M)x), \qquad u = \frac{w}{4} + b(x, y).$$

A direct canonical computation gives the regularized level

$$|u|^2 = F_c(x, y), \tag{3}$$

where

$$F_c(x,y) = 1 - M + \frac{2Ms}{\sqrt{1-4(x^2-y^2)+4s^2}} - 2cs + M^2 s + 4s^3 - 4Ms(x^2-y^2). \tag{4}$$

The symbolic verifier in the release checks the canonical transformation, the completion of squares, and term-by-term identity between (4) and the C++ implementation.

Define

$$K_0(p,q) = \frac{1}{2}|p+b(q)|^2 - \frac{1}{2}F_c(q). \tag{5}$$

The regularized lift is the compact component of $K_0^{-1}(0)$ surrounding the origin.

4 Exact antipodal symplectic shear

Let

$$G_a(p_1,p_2) = ap_1p_2(p_1^2-p_2^2), \qquad a = \frac{1}{2}, \tag{6}$$

and set $R = \nabla G_a$. The map

$$\Phi_a(p,q) = (p, q - R(p)) \tag{7}$$

is a global polynomial diffeomorphism with inverse $(p,Q) \mapsto (p, Q + R(p))$.

Lemma 4.1. *The map Φ_a is symplectic, exact, antipodal, and fixes the origin.*

Proof. Since $DR = \operatorname{Hess} G_a$ is symmetric, the $dp_i \wedge dp_j$ terms cancel in $\Phi_a^* \sum_i dp_i \wedge dq_i$. With the primitive $\lambda = -q \cdot dp$,

$$\Phi_a^*\lambda = \lambda + dG_a.$$

Because G_a is even and homogeneous of degree four, R is odd, hence $\Phi_a(-p,-q) = -\Phi_a(p,q)$. Finally $R(0) = 0$. □

The exactness statement does not identify the radial contact primitive on the image with the original primitive. The precise transport of characteristic dynamics, contact forms, indices, and the antipodal quotient is given in Lemma 8.1.

5 Tangential Hessian criterion

Write $u = p + b(q)$, $P = Db(q)$, and

$$L = \sum_{i=1}^{2} u_i \operatorname{Hess} b_i(q) - \frac{1}{2}\operatorname{Hess} F_c(q).$$

In the variation frame $(\delta u, \delta q)$, the Hessian of K_0 is block diagonal with blocks I and L. If $K_a = K_0 \circ \Phi_a^{-1}$, the shear contributes

$$S = g_{q,1}\operatorname{Hess} R_1 + g_{q,2}\operatorname{Hess} R_2, \qquad g_q = P^T u - \frac{1}{2}\nabla F_c.$$

The full quadratic form is

$$Q = \begin{pmatrix} I+S & -SP \\ -P^TS & L + P^TSP \end{pmatrix}. \tag{8}$$

On $|u|^2 = F_c(q)$ write $u = A(\cos\phi, \sin\phi)$ and $g^2 = F_x^2 + F_y^2$. The code uses the tangent basis

$$w_1 = (-\sin\phi, \cos\phi, 0, 0),$$

$$w_2 = \left(\cos\phi, \sin\phi, \frac{2AF_x}{g^2}, \frac{2AF_y}{g^2}\right), \qquad w_3 = (0, 0, -F_y, F_x).$$

Let $W = (w_1\ w_2\ w_3)$. A direct substitution shows $dK_a(w_j) = 0$ and the matrix tested by the interval code is

$$B_r = W^T Q W. \tag{9}$$

Proposition 5.1. *Suppose the regularized component is smooth, compact, connected, and oriented by the outward normal. If B_r is positive definite at every point, the component is the boundary of a strictly convex body.*

Proof. For a regular level $K_a = 0$, the second fundamental form is

$$\mathrm{II}(v, v) = \frac{\mathrm{Hess}\, K_a(v, v)}{\|\nabla K_a\|}.$$

Thus positive definiteness of (9) is positive definiteness of the second fundamental form. Global strict convexity follows from the Hadamard global convexity theorem in its hypersurface form [9]. □

6 Certified decomposition of the energy range

Figure 1 shows the interval decomposition. The interfaces are exact, so the union is $(c_{L_1}, +\infty)$ with no gaps.

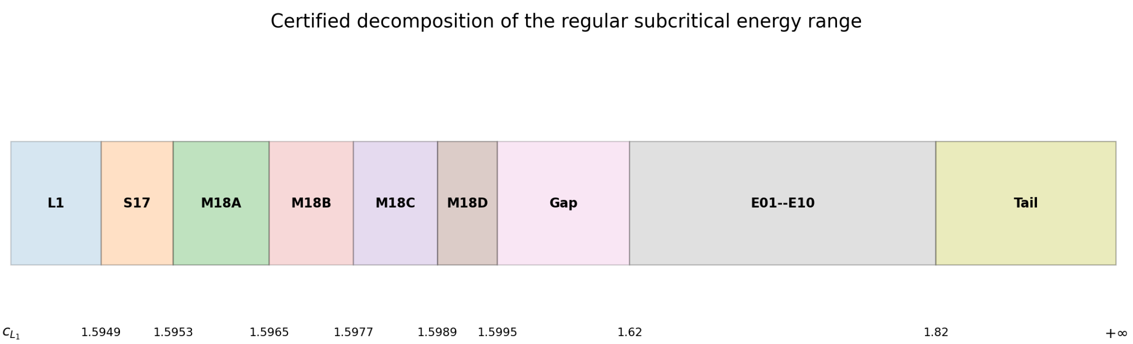

Figure 1: Certified energy decomposition. The horizontal lengths are schematic.

Table 1: Certified blocks and decisive positive lower bounds. Every failed count is zero.

Block	Energy interval	Method	Certified scale	Lower bound
Critical L1 bridge	$(c_{L_1}, 1.5949]$	normal-form + interval bridge	–	0.34012769
S17	$[1.5949, 1.5953]$	adaptive compact atlas	8,192 cells; 567,389 leaves	3.459×10^{-8}
M18A	$[1.5953, 1.5965]$	adaptive compact atlas	8,192 cells; 1,286,974 leaves	1.248×10^{-10}
M18B	$[1.5965, 1.5977]$	adaptive compact atlas	8,192 cells; 1,102,450 leaves	1.802×10^{-8}
M18C	$[1.5977, 1.5989]$	adaptive compact atlas	8,192 cells; 970,092 leaves	2.304×10^{-9}

Block	Energy interval	Method	Certified scale	Lower bound
M18D	$[1.5989, 1.5995]$	adaptive compact atlas	8,192 cells; 850,159 leaves	5.09×10^{-10}
Lower finite gap	$[1.5995, 1.62]$	two cumulative finite atlases	16,384 cells; 6,174,073 leaves	2.047×10^{-9}
Finite Kepler atlas	$[1.62, 1.82]$	ten certified blocks E01–E10	81,920 cells; 2,968,830 leaves	3.739×10^{-9}
Analytic tail	$[1.82, +\infty)$	radial and Hessian interval proof	262,144 xy boxes; 1,048,576 radial boxes	0.52978325

The smallest positive margin in the finite atlas is

$$\boxed{1.2482059830176695 \times 10^{-10}},$$

attained in M18A. Every block has zero terminal failures and every interface is covered from both sides.

6.1 Critical bridge and compact atlas

The interval $(c_{L_1}, 1.5949]$ is not treated by extrapolating the finite Kepler atlas. It is covered by a separate validated normal-form bridge whose twelve promoted stages locate the saddle–center equilibrium, construct the local center branch, transport it to the physical Levi–Civita chart, and verify tangential convexity on the core, boundary collar, and overlap with the compact atlas. The first critical value is enclosed in

$$c_{L_1} \in [1.5941705588746069, 1.5941705588746333].$$

The validated curve reaches

$$c \in [1.5949031618146574, 1.5949031618182739],$$

so its endpoint lies strictly above 1.5949. The decisive Stage 4B bounds are

$$\min \mathrm{Gersh}_{\mathrm{core}} \geq 0.3401276901304223, \qquad m_{\mathrm{bdry}} \geq 1.0521732393534329 \times 10^{-4},$$

$$m_{\mathrm{origin}} \geq 4.865856096238942 \times 10^{-4}, \qquad m_{q,\mathrm{interface}} \geq 2.8035033924207209.$$

At $c = 1.5949$ the critical outer-radius upper bound 0.27471283972361227 lies below the compact-atlas lower radius 0.275, and the corresponding amplitude bounds overlap in the certified orientation. This gives an exact analytic–interval interface; Appendix A records the stage structure and endpoint argument.

The compact interval $[1.5949, 1.5995]$ is then covered by S17 and M18A–M18D. M18A is the narrowest-margin block and therefore the global finite-energy bottleneck. Its full audit verifies 8,192 cells, 1,286,974 accepted leaves, zero failures, and 76,131 manifest entries.

6.2 Lower finite gap

The interval $[1.5995, 1.62]$ is covered by two cumulative finite atlases. Together they contain 16,384 cell-block certificates and 6,174,073 accepted interval leaves, with zero rejected leaves and zero terminal failures.

6.3 Finite Kepler atlas

The interval $[1.62, 1.82]$ is divided into ten blocks E01–E10. Each block contains 8,192 spatial cells and is independently promoted after exact coverage audit and GCC/Clang replay. The final reproduced cumulative master contains 81,920 cells and 2,968,830 accepted interval leaves. An earlier historical master dated 12 July reported 2,963,381 leaves. The difference of 5,449 is caused by operational timeout and subdivision choices in the adaptive tree; the invariant proof gates agree: 81,920 covered cells, zero rejected leaves, zero terminal failures, and the same global worst margin in E07. All numerical statements in this paper use the final reproduced master of 14 July; the reconciliation is archived in `data/KEPLER_COUNT_RECONCILIATION.json`.

6.4 Analytic high-energy tail

For $c \geq 1.82$, the lunar component is confined to $s < 0.026$. The defining function decreases with c at fixed (x, y), while the relevant Hessian block gains a positive contribution as c increases. Hence $c = 1.82$ is the dominant endpoint. A fine interval grid certifies 1,048,576 radial boxes and 262,144 Hessian boxes, with

$$\max F_c(0.026) < -0.00552719716701495, \quad \max \partial_s F_c < -0.6625127849305054,$$

$$\min \det L > 0.3530006021041353, \quad \min \mathrm{Gersh}(L) > 0.5297832479911014.$$

7 Global strict convexity theorem

Theorem 7.1 (Earth–Moon global convexity). *Let $M = 0.987849414390376$, $\mu = 0.012150585609624$, and let c_{L_1} be the first critical value in the convention $\Sigma_c = H^{-1}(-c)$. For every $c > c_{L_1}$, the Levi–Civita lift of the bounded lunar component is carried by an exact antipodal symplectic diffeomorphism to a smooth, compact, connected, strictly convex hypersurface in $\mathbb{R}^4$.*

Proof. The critical bridge and the eight finite/asymptotic regimes cover every $c > c_{L_1}$ with exact interfaces. In each finite regime, the gradient, core, radial, and interface gates identify the same compact lunar component and certify regularity, connectedness, and outward orientation. The interval matrix B_r in (9) is positive definite everywhere; the global minimum certified lower bound is $1.2482059830176695 \times 10^{-10}$. The analytic tail supplies uniform positivity for all $c \geq 1.82$. The preceding proposition therefore implies strict convexity for every regular subcritical energy. □

8 Contact-topological consequence

Let

$$\lambda_0 = \frac{1}{2}\sum_{j=1}^{2}(p_j\, dq_j - q_j\, dp_j), \qquad d\lambda_0 = \omega_0,$$

be the radial Liouville primitive on $\mathbb{R}^4$. The following lemma makes explicit the effect of the shear on the contact dynamics.

Lemma 8.1 (Contact transport through the exact antipodal shear). *Let Σ_0 be the Levi–Civita lift of a bounded lunar component, let D_0 be its bounded domain, and put $\Sigma_a = \Phi_a(\Sigma_0)$ and $D_a = \Phi_a(D_0)$. If Σ_a is strictly convex, then:*

(i) $0 \in \operatorname{int} D_a$, *hence Σ_a is star-shaped with respect to the origin and $\alpha_a = \lambda_0|_{\Sigma_a}$ is a contact form;*

(ii) *the Reeb field of* α_a *spans the Hamiltonian characteristic line of* Σ_a *with its positive orientation;*

(iii) *on* Σ_0,

$$\Phi_a^* \alpha_a = (\lambda_0 - dG_a)|_{\Sigma_0};$$

the associated Reeb trajectories are the original Hamiltonian characteristics up to positive time reparametrization;

(iv) *symplectic conjugacy and positive time reparametrization preserve the Conley–Zehnder indices used in dynamical convexity;*

(v) α_a *is invariant under the antipodal map and descends to the regularized quotient* $\Sigma_a/\{\pm 1\} \simeq \mathbb{R}P^3$.

Proof. The origin lies in D_0 because $K_0(0) < 0$, and $\Phi_a(0) = 0$, so $0 \in D_a$. Strict convexity then implies radial star-shapedness and outward transversality of the radial Liouville vector field. Since $d\alpha_a = \omega_0|_{T\Sigma_a}$, its Reeb line is the positively oriented characteristic line.

For the homogeneous quartic G_a, Euler's identity gives $p \cdot \nabla G_a = 4G_a$. Substitution in the radial primitive yields

$$\Phi_a^* \lambda_0 = \lambda_0 - dG_a.$$

Because Φ_a is symplectic, the kernels of $\omega_0|_{T\Sigma_0}$ and $\Phi_a^*\omega_0|_{T\Sigma_0}$ agree, and normalization by the pulled-back contact form changes only the positive time parameter. The derivative $D\Phi_a$ symplectically conjugates the linearized characteristic flows; the shear isotopy Φ_{ta} connects this conjugacy to the identity, so the relevant Conley–Zehnder indices are unchanged. Finally G_a is even, Φ_a is antipodal, and $(-\mathrm{id})^*\lambda_0 = \lambda_0$, proving invariance and descent to the quotient. □

By the theorem of Hofer–Wysocki–Zehnder, the Reeb flow on a smooth strictly convex hypersurface in $\mathbb{R}^4$ is dynamically convex [10]. Lemma 8.1 transports this conclusion to the original regularized Hamiltonian characteristics and to the antipodal quotient.

Birkhoff's shooting construction gives a retrograde periodic orbit for each regular subcritical energy [1]. Its simple physical projection surrounds the light primary. In the regularized $\mathbb{R}P^3$ component, this orbit is a 2-unknot with rational self-linking number $-1/2$; this is the topology used in Hryniewicz and Salomão [5].

Theorem 8.2 (Earth–Moon disk-like global section)**.** *For every* $c > c_{L_1}$*, Birkhoff's retrograde orbit in the bounded lunar component of the Earth–Moon PCR3BP is the binding of a rational open book decomposition of the regularized* $\mathbb{R}P^3$ *component. Every page is disk-like and is a global surface of section for the Hamiltonian flow.*

Proof. The quotient contact form is dynamically convex by the global convexity theorem and Lemma 8.1. The retrograde orbit is 2-unknotted with rational self-linking number $-1/2$. Corollary 1.8 of Hryniewicz and Salomão [5], with $p = 2$ and $L(2,1) = \mathbb{R}P^3$, gives the rational open book and its disk-like global pages. □

Figure 2 summarizes the proof architecture.

9 Computer-assisted proof architecture

The proof is certificate based. A promoted parent contains source code, exact decimal inputs, raw outputs or compressed ledgers, an independent structural audit, environment metadata, a

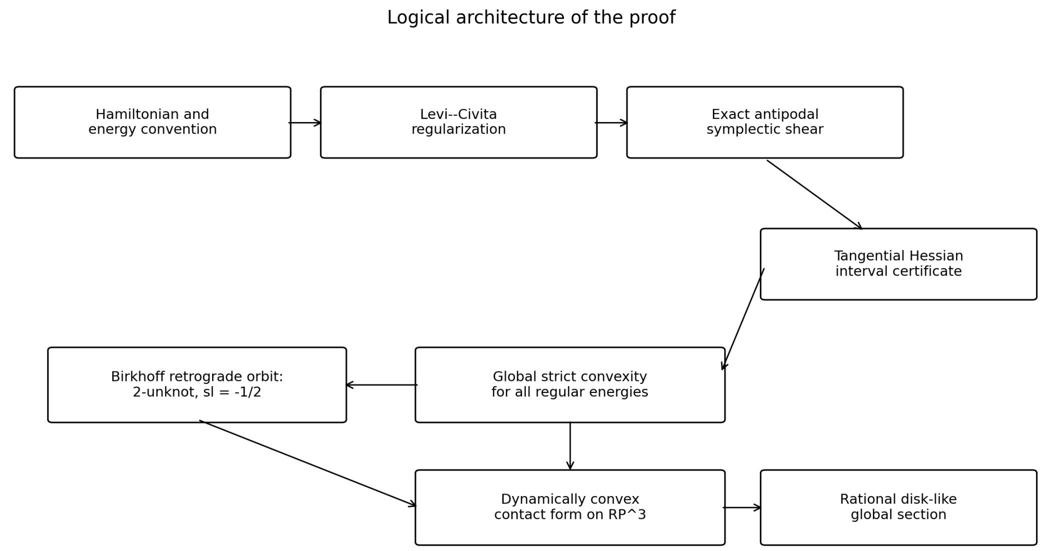

Figure 2: Logical architecture of the proof.

manifest, SHA-256 sums, cross-compiler replay, and a verifier. The thin global master records the exact parent hashes and checks every parent by content hash rather than filename.

The principal finite-energy kernel uses Boost interval arithmetic with outward-rounded transcendental operations. Decimal endpoints are widened by `nextafter` before interval construction. Positive definiteness is certified after midpoint Cholesky preconditioning by a rigorous Gershgorin lower bound, with a raw determinant lower bound retained as a secondary diagnostic. Energy and angular boxes are subdivided adaptively; a leaf is accepted only when every required gate is positive.

A failed exploratory attempt to use the Kepler shear pipeline directly in S17 was rejected after three terminal negative leaves. No result was altered to force acceptance. The critical region was instead certified by the recovered critical pipeline L1/S17/M18A–M18D. This separation is preserved in the release history.

10 Reproducibility

The source repository is organized so that all public identifiers can be inserted after deposition:

- GitHub repository: `xxxxx`;
- Zenodo record: `xxxxx`;
- archive DOI: `xxxxx`;
- software DOI: `xxxxx`;
- version tag: `xxxxx`.

The release includes a Dockerfile, a Conda environment, the exact source kernels, symbolic verifiers, parent-hash registry, M18A reassembly script, release verifier, and one-command audit entry point. The external parents are large and are stored in the archival data release; the article repository keeps their immutable SHA-256 identifiers.

11 Discussion

The proof covers the physical Earth–Moon mass ratio over the full regular subcritical range rather than a perturbative mass regime, a local energy interval, or the Hill limit. The use of a global exact shear is essential in the finite-energy range, whereas the identity map suffices in the analytic tail. The narrow M18A margin explains why a single coarse atlas or an unmodified Kepler pipeline is inadequate near the critical regime.

The disk-like global section reduces the flow to an area-preserving return map on a disk. This creates a rigorous foundation for subsequent work on direct periodic orbits, rotation theory, symbolic dynamics, transport, and computer-assisted analysis of the return map. A complementary route for parameter regions where no convexifying embedding exists is the direct interval certification of Conley–Zehnder indices, for which a validated pipeline was developed in Grisa [11].

Data and code availability

All source code, certificates, logs, manifests, environment files, and verification scripts are available at GitHub `xxxxx` and Zenodo `xxxxx`. The archival DOI is `xxxxx`.

Author contributions

César Augusto Grisa: conceptualization, mathematical analysis, computational design, software, validation, data curation, visualization, and writing.

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Competing interests

The author declares no competing interests.

A Validated critical L_1 bridge

The critical bridge is a separately promoted analytic–interval proof, not a numerical continuation of the compact atlas. Its twelve stages isolate the L_1 saddle–center equilibrium, validate the local center branch in normal-form variables, transport it through exact coordinate changes to the Levi–Civita chart, and certify the core, analytic collar, origin, and compact-atlas overlap.

The curve

$$w(s) = (s, 0, -s, 0), \qquad 0 \leq s \leq 0.0158,$$

is validated by Taylor–defect enclosures of degrees 12 and 16. Its endpoint energy lies in

$$[1.5949031618146574, 1.5949031618182739],$$

whose lower endpoint exceeds 1.5949 by $3.1618146572043315 \times 10^{-6}$. Stage 4B further certifies

$$D_{-}^{\text{outer}} \leq -0.056262049365230826, \qquad F_{-}^{\text{outer}} \leq -0.001049253821527877,$$

together with the positive margins listed in Section 6.1. Convexity of the transverse boundary expression reduces its maximum to the certified endpoints. The endpoint identities and an intermediate-value argument along the validated center branch cover every energy in $(c_{L_1}, 1.5949]$. GCC/Clang replay, a higher-precision rerun, and the symbolic identities all return PASS. The overlap with the compact atlas is certified independently at $c = 1.5949$, so no limiting extrapolation is used.

B Exact artifact registry

L1 critical bridge `PCR3BP_EARTHMOON_L1_CRITICAL_BRIDGE_MASTER_20260711.zip`
SHA-256: `a6666bd89d0027022b0ad68cd82ff91055217e1bbbc94d305523156b9938604d`

S17 `PCR3BP_EARTHMOON_FULL_SUBSLAB_S17_20260710.zip`
SHA-256: `e67e509e7208646d51c0127f6463e2bd29596f44cb7c384017da8c6ddd017724`

M18A `PCR3BP_EARTHMOON_ADAPTIVE_MACROBLOCK_M18A_REBUILT_20260711.zip`
SHA-256: `35b6e116789700f87b7ad80053d5eeeab964e65e12bc3a17a3402919990f0bd4`

General compact master `PCR3BP_EARTHMOON_GENERAL_CONTINUITY_MASTER_20260712.zip`
SHA-256: `2e34445f7728d8609019487236bb6cebe6db16bc5ff26bee6c6c013d3d379016`

Lower gap `PCR3BP_EARTHMOON_LOWER_GAP_COMPLETE_MASTER_15995_162_20260713.zip`
SHA-256: `ac2d5d85b8627a2a1c4b653d0826070cc1bf15f277d791ff895e743918fe92e5`

Kepler/high energy `PCR3BP_EARTHMOON_KEPLER_COVERAGE_1P62_TO_INFINITY_MASTER_REPRODUCED_20260714.zip`
SHA-256: `ad20cd26697e3c705e303ede17fd949125040d0c403e81eccd21fbc36f0820bf`

Theorem mapping `PCR3BP_EARTHMOON_THEOREM_MAPPING_T1_T7_CHECKPOINT_20260713.zip`
SHA-256: `1bc0aa1199f01bcf62910844e2d8a211801eac68e9e054f28b6586c2fd4b3308`

Shear/Hessian audit `PCR3BP_EARTHMOON_SHEAR_CONTACT_HESSIAN_AUDIT_CHECKPOINT_20260713.zip`
SHA-256: `dd68d0f7c3f1645adb0044813842e22019767fc75ea0a804931d0a664f25273e`

Topology reduction `PCR3BP_EARTHMOON_TOPOLOGY_REDUCTION_CHECKPOINT_20260713.zip`
SHA-256: `155e5e1f1ffa40627f843d7d28801921e6b14b8e1fe8f66bfd10ec85f3df6b29`

Theorem assembly `PCR3BP_EARTHMOON_CONDITIONAL_GLOBAL_THEOREM_ASSEMBLY_CHECKPOINT_20260713.zip`
SHA-256: `3cae8ac0580a0f0e71c4c13879fe420c100ea5ef0e23280683a6a81f67d8bb14`

C Release verification commands

```
make verify
make verify-parents PARENTS=/path/to/zenodo-release
make manuscript
make package
```

The release verifier refuses missing files, size mismatches, hash mismatches, failed CRC checks, non-PASS parent statuses, interval gaps, rejected leaves, and terminal failures.

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